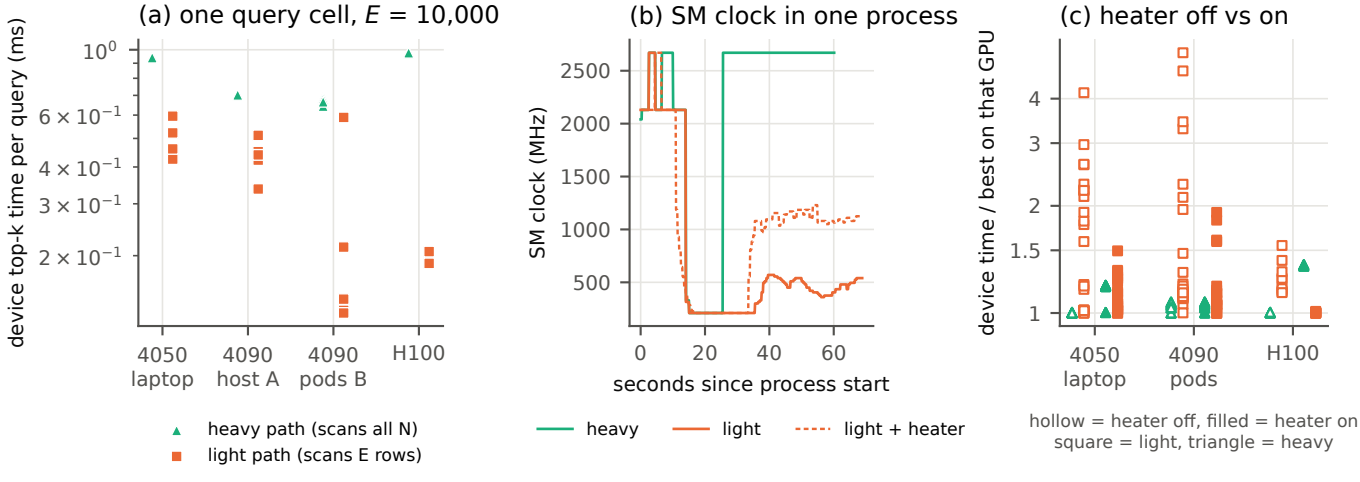

(a) one query cell, E = 10,000
device top-k time per query (ms)
10^0
6×10^{-1}
4×10^{-1}
3×10^{-1}
2×10^{-1}
4050 laptop
4090 host A
4090 pods B
H100
heavy path (scans all N)
light path (scans E rows)
(b) SM clock in one process
SM clock (MHz)
2500
2000
1500
1000
500
0
20
40
60
seconds since process start
heavy
light
light + heater
(c) heater off vs on
device time / best on that GPU
4
3
2
1.5
1
4050 laptop
4090 pods
H100
hollow = heater off, filled = heater on
square = light, triangle = heavy